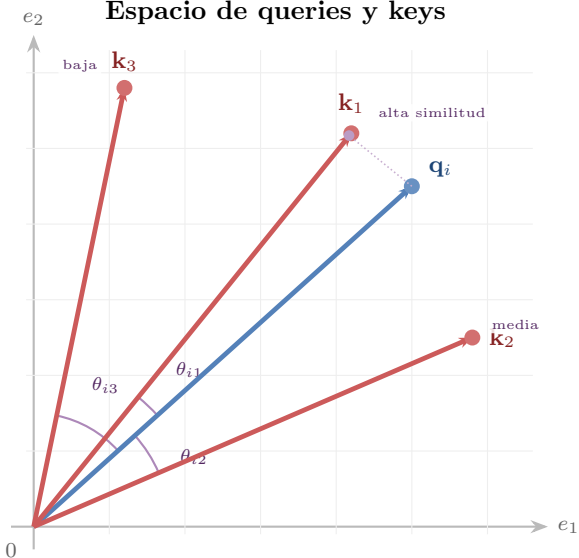


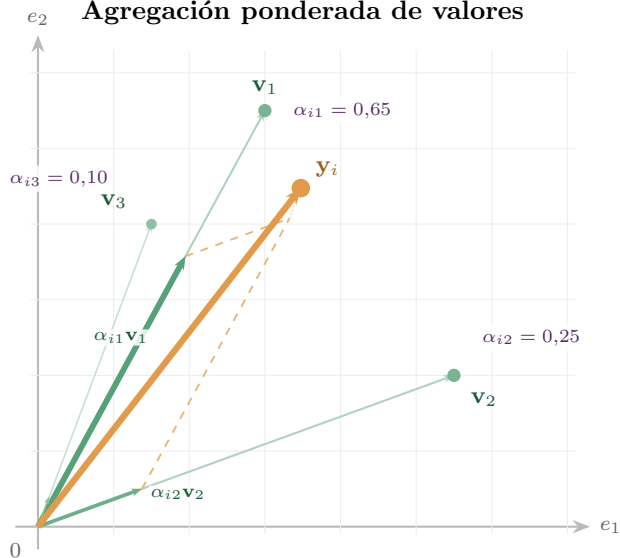
## Espacio de queries y keys



$$\alpha_{ij} = \text{softmax}_j \left( \frac{q_i \cdot k_j}{\sqrt{d_k}} \right)$$

## Agregación ponderada de valores

Pesos de atención  $\alpha_{ij}$



$$y_i = \sum_j \alpha_{ij} v_j$$

→ Query  $q_i = W_Q x_i$

→ Keys  $k_j = W_K x_j$

→ Values  $v_j = W_V x_j$

→ Salida  $y_i$